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J. Appl. Phys. 133, 015703 (2023)
https://doi.org/10.1063/5.0124285

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Cite as: J. Appl. Phys. **133**, 015703 (2023); doi: 10.1063/5.0124285
Submitted: 4 September 2022 · Accepted: 13 December 2022 ·
Published Online: 4 January 2023

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ABSTRACT

Gallium oxide is an emerging wide-bandgap semiconductor with promise for applications in space systems that may be exposed to energetic particles. We use molecular dynamics simulations, based on *first principles* density-functional methods, to determine the nature and stability of the defects generated by atoms knocked-out by particle irradiation at near threshold energies (found to be 28 ± 1 eV for Ga and 14 ± 1 eV for O). For Ga atoms, several types of low energy knock-out events result in defect complexes, but the final structures depend critically on the initial displacement direction. In contrast, a vacancy plus a peroxide linkage occurs in all types of low energy knock-out events of O atoms. Based on energy-barrier calculations, there is a low (high) probability for Ga (O) defect recombination. The electronic structure of residual, relaxed defects generated by Ga knock-outs reveals defect levels near the band edges.

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I. INTRODUCTION

The β phase of gallium oxide (β-Ga_2O_3) is an ultra-wide-bandgap semiconductor ($E_{gap} \sim 4.9$ eV) that can be grown in high-quality crystals. Controllable n-type conductivity can be realized by the ion implantation of several elements (e.g., Si or Ta) as shallow donors or by elements such as Mg as deep acceptors compensating the as-grown n-type conductivity.[1,2] The material has been incorporated in many interesting electrical devices including as UV transparent conducting contacts for solar cells and as the base semiconductor for high-power and displacement-damage resistant field-effect transistors.[2,3] The effects of radiation-induced displacement damage on gallium oxide have recently been the subject of numerous studies examining the effects of neutron,[4,5] electron,[6] proton,[7,8] and heavy-ion[9] bombardment on the properties of gallium oxide materials and devices.[3] Understanding the defect creation processes caused by high-energy particles is important for developing gallium oxide for applications in electronics.

Point defects in as-grown gallium oxide have been the subject of intense study for over a decade.[10,11] The established understanding of point defects has resulted from the culmination of several experimental and theoretical studies; for a detailed review, see the article by McCluskey.[11] Recent work has combined theory and experimental data to examine the point defects left over after the proton irradiation of gallium oxide,[7,8] where defects were observed with electron paramagnetic resonance (EPR)[7] and deep-level transient spectroscopy (DLTS).[8] Density-functional-theory (DFT) calculations led to the suggestion that isolated gallium vacancies cannot explain all the experimental results but that vacancy complexes may.[7,8]

The range and concentration of point defects in materials caused by irradiation or ion implantation depends on the threshold

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energy (E_{th}), i.e., the minimum energy needed to knock out a single atom from its ideal lattice site. During implantation or irradiation, particles with energy up to millions of eV penetrate the material and interact with lattice atoms, causing numerous knock-out events. These primary knock-out events can lead to secondary knock-outs, resulting in a defect cascade. To calculate the properties of defect cascades, numerous studies[7–9] use the SRIM package,[12] which uses E_{th} to determine the effects of particle–atom collisions in bulk and heterojunction systems within a Monte Carlo simulation. Based on an empirical relation between E_{th}, material density, and bandgap,[3,12] the threshold energy for Ga in gallium oxide (E_{th}^{Ga}) is estimated to be 25 eV,[7,9] a value which is slightly larger than the experimentally derived E_{th}^{Ga} of 20 eV in GaN.[13,14] For O threshold energies (E_{th}^{O}) in gallium oxide, recent studies use 28 eV.[7,9] However, in recent threshold calculations of oxygens in SiO_2 [15] and Al_2O_3,[16,17] the oxygen threshold is found to be ∼24 eV lower than either the Si or the Al threshold energies, respectively. Therefore, we expect the lighter O atom to have an appreciably lower threshold than Ga atoms in Ga_2O_3.

It is important to determine E_{th} experimentally, while understanding the relevant processes using first principles simulations. The first principles calculations can provide important details of low energy knock-out events including the nature and stability of the generated defects. First principles simulations have been successfully employed to examine knock-out events in several semiconductors including SiC,[18] Al_2O_3,[16] GaN,[19] SiO_2,[20] and AlN.[21] However, to the best of the authors' knowledge, neither *first principles* nor empirical-potential MD simulations have been performed to examine threshold energies and defect-formation processes for knock-out events in Ga_2O_3.

In this paper, we present the results of extensive *first principles* MD simulations that monitor the trajectory and final resting point of individual Ga and O atoms that are given a range of initial kinetic energies with initial velocities in different directions. These starting conditions represent atoms that are displaced by incident high-energy irradiation. If the ejected atom returns to its original site, the simulation is repeated until a minimum initial kinetic energy is identified that results in the atom coming to rest in some configuration other than its original site. A large number of distinct directions are investigated to determine the overall E_{th} for each of the two atomic species. The atomic and electronic structures of the stable defects that are present at the end of the ballistic trajectory of low energy knock-out events, as well as the energy barriers for knocked-out atoms to return from their rest site to their original site, are determined. There is a low probability for Ga atoms to return to their initial site (defect recombination), while the probability of O atoms to do so is high. Electronic structure calculations reveal that residual, relaxed Ga defects have energy levels near the band edges. The present study fills a gap in the present understanding of radiation effects in Ga_2O_3 and provides a basis for further work to simulate defect cascades.

II. METHODS

To determine the threshold energies from MD simulations, we use the Vienna First principles Simulation Package (VASP[22]) that is based on DFT.[23,24] We treat exchange-correlation effects with the semi-local Perdew–Burke–Ernzerhof (PBE) functional.[25] Standard projector-augmented-wave (PAW) potentials are used to represent the core-electrons.[26,27] For gallium, the three outer valence electrons ($4s^2\ 4p^1$) and, for oxygen, the eight outer valence electrons ($2s^2\ 2p^6$) were explicitly included in the DFT calculations. For the plane wave basis used to expand wave functions, the cutoff energy was set at 400 eV. These choices lead to the converged results based on the tests reported in Fig. S1 in the supplementary material using "harder" PAW potentials that include explicitly more atomic electrons in the valence set (Ga $3d^{10}$) and using a plane wave cutoff energy of 600 eV. Overall, we estimate a ±1 eV for our reported threshold energies.

The gallium oxide unit cell is monoclinic with the theoretical lattice parameters of $(a, b, c) = (12.45, 3.082, 5.877)$ Å and $(\alpha, \beta, \gamma) = (90°, 103.7°, 90°)$, which are close to previously found values.[10] The supercell used is $1 \times 5 \times 3$ multiples of the unit cell and includes 300 atoms. The use of 300 atom supercells provides enough atoms in the system to accurately simulate the knock-out events. Since these are low energy events, Ga and O knock-out atoms move less than half a nano-meter and stay well within the supercell. Using larger supercells with many more atoms would allow us to examine larger energy knock-out events but would make the computational costs prohibitive.

Given the size of the supercell, only the gamma point is used for Brillouin-zone integrations. Also, the lattice parameters are fixed during the MD simulations. To determine the threshold energy from ion–lattice collisions, we follow a procedure similar to previous *first principles* MD studies.[19–21] Initially, the atoms are relaxed, so the system temperature is at T = 0. At room temperature, atoms have on average only ≈0.025 eV of kinetic energy. Since $E_{th} \gg 0.025$ eV, lattice vibrations at room temperature are not expected to impact the threshold values found in this study.

To search for the threshold energies of gallium oxide, a primary knock-on atom is selected along with a direction for its given velocity. The kinetic energy of the knock-on atom is initially set at a low value: 20 eV for Ga and 10 eV for N. All atoms evolve according to Newtonian dynamics with forces calculated by quantum mechanics using VASP and a time step of 1 fs. With this time step choice, the knock-on atom moves at most $\Delta r_{max} < 0.2$ Å between force calculations for the largest initial kinetic energies considered, but $\Delta r_{max} < 0.1$ Å for initial kinetic energies near the threshold values. After the MD simulation runs for 1000 fs, all atoms are relaxed using a conjugate-gradient-minimization procedure. The atomic positions are relaxed until forces are all less 0.03 eV/Å. If the system relaxes back to the original crystalline arrangement, then the initial knock-on atom's kinetic energy is raised by 5 eV until a defected system emerges from the final relaxation step. Then, the initial kinetic energy of the knock-on atom is changed by 1 eV until the threshold energy is found.

We also performed a test to ensure the threshold-energy results are robust. Each threshold energy is found for a particular knock-on atom in a particular direction from the 1000 fs MD simulation. Then, a second MD simulation is performed for 2000 fs. In some test cases, the time step was reduced to 0.66 fs, but the lower time step had no effect on the final threshold-energy results. Only when defect formation occurred after both a 1000 fs simulation and a 2000 fs simulation was a knock-out event recorded. Below, we

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report the final structures and energies resulting from the longer 2000 fs MD simulations. Overall, for a particular atom in a particular direction, we estimate a ±1 eV numerical uncertainty for the reported threshold energies.

Defect-formation energies were calculated as if the defects were formed under equilibrium conditions using standard formulas.[28] Such formation energies only give us a rough idea of the relative energy cost of different defects that are generated by the kick-out processes. Energy barriers for defect rearrangements were calculated using the nudged-elastic-band (NEB) method.[29]

For all the above calculations, including all defect-structure relaxations, we employed the PBE exchange-correlation functional, but this functional is not suitable for the calculations of defect energy levels in the energy gap because the PBE-calculated energy gap is only about half the experimental value. Therefore, for the electronic structure of defects, we employed a single electronic convergence calculation using the HSE hybrid-exchange method[30,31] with a PBE-exact-exchange mixing parameter of 0.35 and a screening parameter of 0.12 $Å^{-1}$. This HSE method results in a bandgap of 4.9 eV, which is in agreement with experiments on bulk Ga_2O_3 so that the calculated defect energy levels are correctly positioned in the energy gap. Given the size of the supercell, a single k-point at Γ is used for Brillouin-zone integrations for defect energy-level calculations. We calculated the electronic density of states (DOS) using a Gaussian broadening of 0.05 eV for each state.

III. RESULTS

Previous threshold-energy studies have found that low energy knock-out events occur along directions that avoid nearest-neighbor atoms and point toward open interstitial regions.[19–21] In gallium oxide, there are two types of Ga sites and three O sites. The gallium atom forms four Ga–O bonds at site I, while it forms six Ga–O bonds at site II. The oxygen atoms at sites I and II each have three Ga–O bonds, whereas the atoms at site III have six Ga–O bonds. Knock-on directions resulting in low threshold energies are illustrated in Fig. 1 for each type of atom in gallium oxide. Figure 1 shows the *a-c plane* of the supercell, but none of the Ga–O bonds are in this plane. For O(I), the (001) direction appears to point directly at the nearest Ga atom, but the line actually passes through an interstitial region ∼1.5 Å from the two nearest Ga atoms. Similar analysis applies for Ga(II) in the (00-1) direction and O(II) in the (100) direction.

Following the procedure outlined in Sec. II, we calculated the threshold energies along dozens of directions. For clarity, in Tables I and II, we report only the five lowest-threshold-energy direction results for each type of atom considered. See Table SI in the supplemental material for the full list of results. Initially, we explored the simplest promising directions such as (101), (111), and (001). After the lowest-threshold-energy directions were identified, we considered variations that seemed promising, such as (1 0 −3) and (3 0 −1).

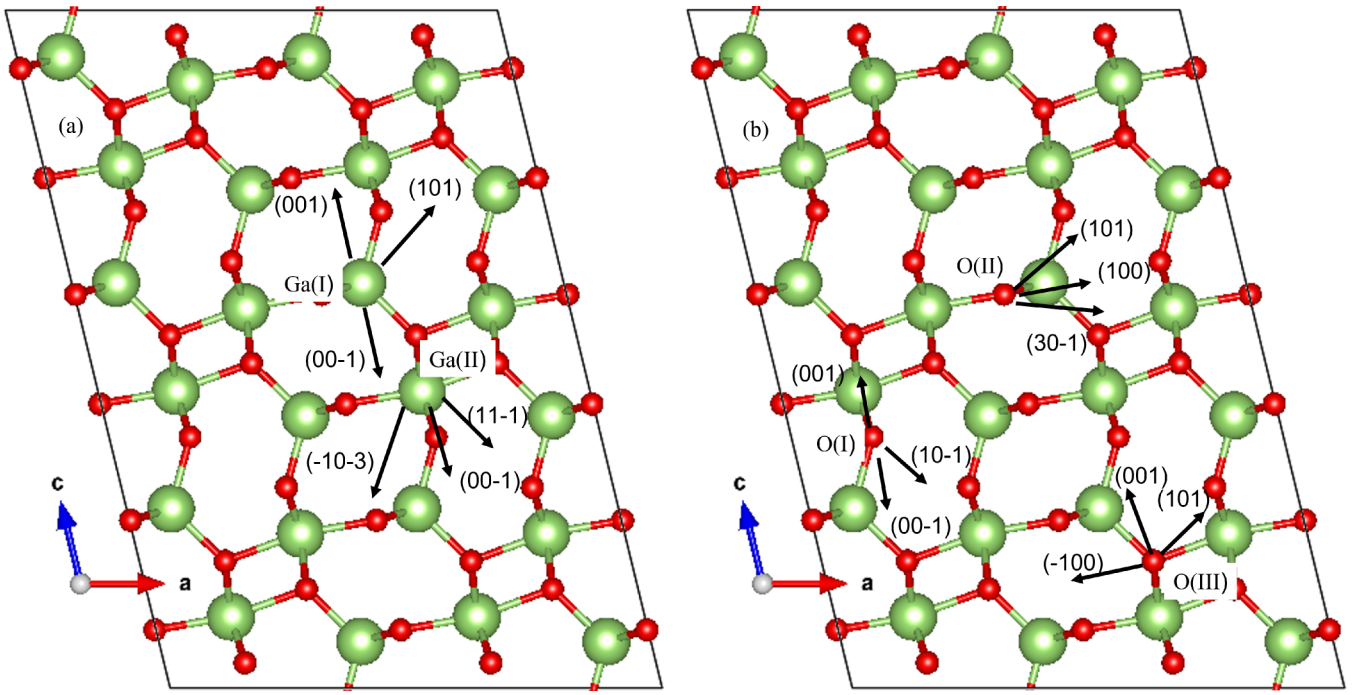


FIG. 1. Ball-and-stick models of the local bonding environments in Ga_2O_3 identifying the knock-on atom directions that lead to low energy knock-out events for the various types of atoms: (a) Ga(I), (b) Ga(II), (c) O(I), (d) O(II), and (e) O(III). The length of the vectors is chosen simply to maximize the visibility of the vectors in the model. The lattice vectors are also included. The directions of the initial velocity vectors are in the Cartesian system. The lattice vectors are related to Cartesian vectors as follows: $\hat{a} \cong 0.97\,\hat{x} - 0.24\,\hat{z}$, $\hat{b} = \hat{y}$, $\hat{c} = \hat{z}$.

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TABLE I. For each type of atom, the table reports the threshold energy (E_{th}) for a given knock-on direction along with the Frenkel-pair separation (ΔR). Rows with highlighted * are the minimum energy knock-out directions described in more detail in Table II.

Atom	Site	Direction	E_{th} (eV)	ΔR (Å)
Ga	I	1 0 1	31	3.6
	I	1 1 1	34	5.5
	I	0 0 1	34	4.4
	I	0 0 −1	34	4.2
	I	−1 1 1	34	4.1
Ga	II	0 0 −1	28*	3.9
	II	−1 0 −3	29	4.9
	II	1 1 1	30	2.6
	II	1 1 −1	30	2.6
	II	0 0 1	33	5.0
O	I	0 1 3	15	5.5
	I	0 0 −1	17	11.2
	I	0 0 1	18	11.2
	I	1 0 −1	21	4.1
	I	1 1 1	21	5.5
O	II	3 0 −1	14*	3.6
	II	1 0 0	16	3.6
	II	1 0 1	17	4.1
	II	1 0 −1	18	3.5
	II	1 1 1	18	3.2
O	III	1 1 1	14*	4.4
	III	1 0 1	14*	2.5
	III	1 0 3	15	3.1
	III	0 1 1	15	4.4
	III	0 1 3	16	3.1

For Ga, we found a dozen knock-on directions that result in threshold energies ranging between 28 and 34 eV, with an average threshold energy of 32.3 eV. Several directions for Ga(II) lead to threshold values below the lowest value of 31 eV found for Ga(I). It is noteworthy that compared to the Ga(I) vacancy, the neutral Ga (II) vacancy formation energy is ∼1 eV higher.[32] The lowest–threshold-energy of 28 eV was found for a Ga(II) atom with a knock-on direction of (00-1). The final relaxed defect complex illustrated in Fig. 2(a) includes a gallium vacancy (V_{Ga})–interstitial (Ga_i) Frenkel pair involving the primary knock-on atom, which is joined by two neighboring gallium atoms that shift from their ideal lattice positions. Overall, the defect can be considered a complex of three closely associated Frenkel pairs. After relaxation, the primary knock-on gallium comes to rest, resulting in a Frenkel-pair distance of $\Delta R = 3.8$ Å, which is in the mid-range of the distances observed in this study (2.6 Å $< \Delta R_{Ga} <$ 5.5 Å). The two secondary knock-out events result in two Ga atoms displaced by modest amounts (1.3 and 2.1 Å). The broad ranges of Frenkel-pair distances and defect-formation energies are indicative of the complex range of defects resulting from low energy knock-out events. The formation energies and a fuller description of each Ga defect complex is given in Table S1 and Fig. S2 in the supplementary material.

TABLE II. For the lowest knock-out energies for each type of atom, the table reports the threshold energy (E_{th}) for a given knock-on direction along with the number of Frenkel pairs along with interstitial-vacancy separation (ΔR).

Atom	Site	Direction	E_{th} (eV)	Frenkel pair number [ΔR(Å)]
Ga	II	0 0 −1	28	3, [3.9, 1.3, 2.1]
O	II	3 0 −1	14	1, [3.6]
O	III	1 1 1	14	1, [4.4]
O	III	1 0 1	14	1, [2.5]

For O, we found 20 knock-on directions that lead to threshold values between 14 and 22 eV, with an average threshold energy of 17.7 eV. Several knock-on directions for O(III) lead to threshold values below 16 eV, whereas for O(I) and O(II), there is only one low energy knock-out direction. It is noteworthy that the neutral O(III)-vacancy formation energy is >1 eV larger than the O(I) or O(II) vacancy values.[10] We found three cases where the threshold energy is 14 eV. Here, we focus on the O(III) atom when knocked-out in the (111) direction, which results in the defect complex illustrated in Fig. 2(b). This defect has an oxygen Frenkel-pair distance $\Delta R = 4.4$ Å, which is one of the largest found in this study. Such a large Frenkel-pair distance indicates that it is likely a difficult defect complex to heal. Mostly, all oxygen defects involve a single oxygen vacancy (V_O)–interstitial (O_i) Frenkel pair involving the primary knock-on O atom. In the case shown in Fig. 2(b), the interstitial oxygen forms a peroxide linkage with an O(I) atom where the O–O bond length is 1.5 Å, same as in peroxides generally. The formation energies and a fuller description of each Ga defect complex are given in Table S1 and Fig. S2 in the supplementary material.

In addition to the primary knock-out events discussed above, we also observed that low energy Ga knock-on events in many cases lead to secondary knock-out events, some of which result in oxygen Frenkel-pair defects. The initial knock-on Ga atom returns to its initial lattice site and a secondary oxygen knock-out event occurs in four cases with Ga knock-on energies between 26 and 29 eV. The Frenkel-pair distances and formation energies are all within the range observed in the case of primary O atom knock-out events reported in Table I and also in Table S1 in the supplementary material.

For oxygen knock-out events, the largest resulting Frenkel-pair distance reported in Table I is 11.2 Å found for O(I) atoms with knock-on directions in the (001) or (00-1) directions. In these two cases, direct (billiard ball style) collisions occur with the primary atom knocking out a secondary atom and coming to rest at the secondary atom's crystalline site. In these cascades, four oxygens in total are involved (see Fig. S2 in the supplementary material for pictures of the final defects). The large size of the Frenkel-pair distance in these two cases is likely due to the combination of T = 0 K simulations and the knock-on direction pointing directly to the nearest neighbor. To test this hypothesis, we examined the O(I) atom knocked in the (013) direction, which is slightly off from the (001) direction. Interestingly, we find a low knock-out threshold energy of 15 eV and a large Frenkel-pair distance of 5.5 Å. The final defect structure in this case is presented in Fig. 2(c). Despite the large Frenkel-pair distance, analysis discussed below shows that this defect is likely to heal.

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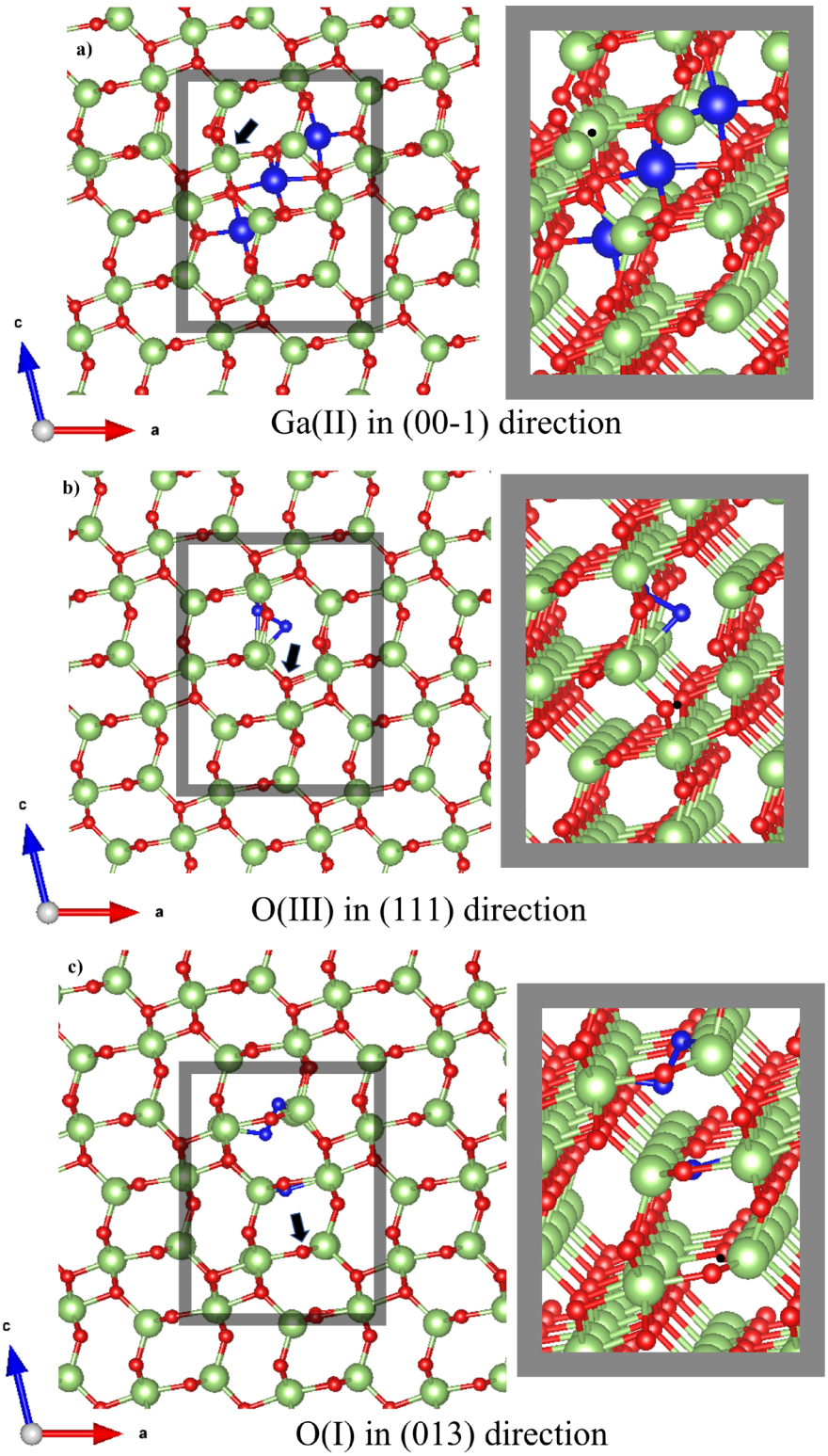


FIG. 2. Ball-and-stick representation[33] of defects resulting from (a) the Ga(II) atom knocked-out in the (00-1) direction with initially 28 eV, (b) the O(III) atom knocked-out in the (111) direction with 14 eV, and (c) the O(I) atom knocked-out in the (013) direction with 15 eV. In all figures, the right image is a slightly titled version of the structure imaged in the shaded box on the left. Also, the arrow on the left and the solid dot on the right indicate the location of the vacancy. Green balls are gallium, and red balls are oxygen. The primary and secondary knock-out atoms are all colored blue.

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IV. ANALYSIS AND DISCUSSION

The rate or probability of defect healing is important for understanding the damage processes during ion bombardment or irradiation. To estimate the probability of healing, we have examined the energy barriers for defect complexes to heal. For the Ga knock-out defects, multiple Ga atoms and numerous Ga–O bonds need to be rearranged. For oxygen defects, the process is much simpler since a single Frenkel defect pair is involved. In this analysis, we consider the Ga defect formed by the Ga(II) atom being knocked-out in the (00-1), which is illustrated in Fig. 2(a). For the O defect, we consider the defects depicted in Figs. 2(b) and 2(c). For each defect, we calculated the barrier for the primary defect atom to hop away and/or toward the accompanying vacancy.

For the Ga defect complex, we attempted to find the barrier for the defect complex and surrounding atoms to return to their pristine coordinates. A converged barrier result was not obtained. Nevertheless, our calculations for a single hopping process place the barrier above 7 eV, suggesting that a multi-step process is more likely. Overall, the Ga defect complex appears difficult to heal.

For O Frenkel-pair defects found, the final peroxide bond is found to be locally stable in all cases. For the defect shown in Fig. 2(b), when the surrounding atoms are allowed to move and relax (which occurs on a nano-second time scale), the barrier for recombination is relatively small (∼0.2 eV). In addition, the barrier perpendicular to the (010) direction is large ∼3.3 eV. Therefore, the interstitial oxygen is confined, diffusing back and forth along the (010) and (0-10) directions through the relatively open channel. At typical experimental/device operation temperatures, the oxygen diffusion would stop if the oxygen interstitial recombines with a vacancy since over 6 eV of energy would be required to recreate the Frenkel pair (see Table S1 in the supplementary material for defect-formation energies). Despite the large Frenkel-pair distance of the defect in Fig. 2(c), the barrier for recombination is small (< 0.4 eV), whereas the barrier for the interstitial oxygen to move away from the vacancy is larger (> 1 eV). See Figs. S3 and S4 in the supplementary material for the specific oxygen barrier results. Given the simplicity of the oxygen defect and the relatively low barrier for recombination, the probability for oxygen Frenkel-pair recombination is high.

Ga defects are likely the most enduring defects created during low energy knock-on events. We examine the electronic structure of these defects in more detail. For these calculations, we use the final atomic positions obtained from the relaxation calculations. The results for the bulk crystal and for the Ga defect shown in the schematic of Fig. 2(a) are reported in Fig. 3 for the near gap DOS. The full DOS plots (not shown) indicate that the energy bands are properly aligned. For the Ga defect complex, two distinct defect peaks form at 0.5 and 0.9 eV above the valence-band maximum. Also, localized band tail states occur at both valence and conduction band edges.

To further understand the nature of the Ga defect complex, the defect eigenstate charge densities were calculated. For the pristine crystal, the valence and conduction band edge states are the delocalized states of oxygen p- and s-character, respectively, as illustrated in Fig. S5 in the supplementary material. In Fig. 4, we report the iso-surface charge density superimposed on the ball-and-stick image from Fig. 2(a). Specifically, the defect state at $E_{VBM} + 0.9$ eV is presented in Fig. 4. The yellow coloring represents regions where the charge density is high. Note that the charge is localized mainly on an oxygen atom directly above the knocked-out Ga Frenkel defect. This charge localization indicates that the defect has a doubly occupied p-state localized on a single O atom, pointing to the Ga vacancy site. There is minimal charge on the defected Ga atoms. Both defect states near the valence-band maximum can be considered oxygen dangling bonds. The conduction band tail states show an extended component and a component localized on both defective Ga and O atoms (see Fig. S6 in the supplementary material).

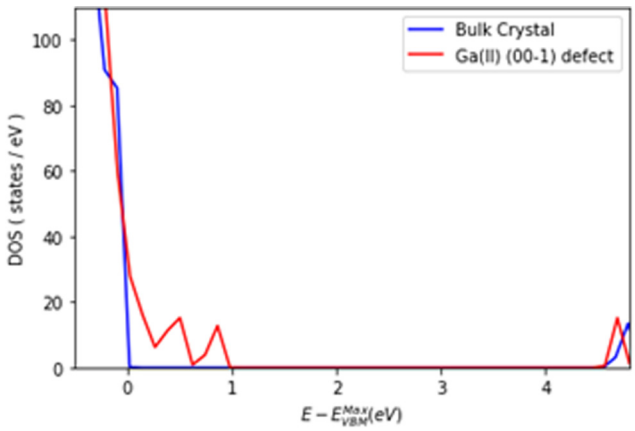


FIG. 3. Showing the DOS vs energy for the bulk crystal and the model from Fig. 2(a).

The localized electron density on the defect atoms could be visible in the EPR experiments. However, the final separation distances and the final charge states of the initial Frenkel-pair clusters will be important to the experimental signatures observed. It is noteworthy that recent EPR experiments that found single Ga vacancy defects are not consistent with observations but clusters

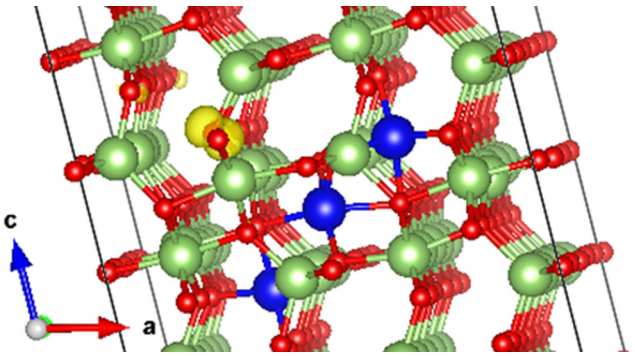


FIG. 4. The iso-surfaces for the eigenstate (yellow) of the defect level at 0.9 eV above the valence band from Fig. 3.

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may be.[7,8] Our simulations indicate that defect clusters should be common since even the lowest energy Ga knock-out events result in clustered defects.

V. CONCLUSIONS

Using dozens of *first principles* molecular dynamics simulations, knock-on events are investigated for Ga and O atoms in Ga_2O_3. The knock-on energy and directions are varied in such a manner to search for the overall threshold energy for displacement damage. For gallium, the Ga(II) atom knocked-out in the (00-1) direction leads to the overall lowest knock-out value of 28 ± 1 eV. The resulting defect appears to be a complex of three Frenkel pairs, which is stable to recombination. The defect complex features two occupied localized states above the valence-band edge. For oxygen, several atom types/directions lead to a threshold energy of 14 ± 1 eV. In all oxygen cases, a single Frenkel pair is formed. For the O(II) atom in the (111) direction, the Frenkel-pair distance is greater than 4 Å, which is the largest of the low energy knock-out events for oxygens. However, the barrier for diffusion is small (~0.2 eV) and limited along one channel so that Frenkel-pair recombination is likely for oxygen defects. For the simulations of ion bombardment[12] of Ga_2O_3, threshold energies are important input parameters. Based on the analysis of our theoretical results, 28 ± 1 eV is the best present estimate of the threshold energy for gallium knock-outs. Given the high probability for oxygen defect recombination, the 14 ± 1 eV value found for the oxygen threshold energy may be lower than the actual threshold for generating observable concentrations of oxygen defects during irradiation experiments. For both Ga and O atoms, all low energy knock-out events involve a significant (001) knock-on component, suggesting that irradiation in this crystallographic direction may be the most damaging.

SUPPLEMENTARY MATERIAL

See the supplementary material for additional information supporting the results and analysis presented above. There is a full presentation of the results of all knock-out simulations performed. In addition, there are calculations reported to provide an understanding of the dynamic and electronic properties of low energy defects.

ACKNOWLEDGMENTS

This work was supported by the Air Force Office of Scientific Research Center of Excellence on Radiation Effects at Vanderbilt University through Grant No. FA9550-22-1-0012 and by the McMinn Endowment at Vanderbilt University. B.R.T. acknowledges the partial support from the Air Force Office of Scientific Research under Award No. FA9550-22-1-0308. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the United States Air Force. Computations for this research were performed on the Department of Defense High Performance Supercomputers using Subproject Nos. AFOSR49463GAN and AFSNW44943FRE.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Blair R. Tuttle: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Nathaniel J. Karom:** Formal analysis (supporting); Visualization (supporting). **Andrew O'Hara:** Formal analysis (supporting). **Ronald D. Schrimpf:** Writing – review & editing (supporting). **Sokrates T. Pantelides:** Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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